CHAPTER 6

POINT-SET TOPOLOGY IN O-MINIMAL STRUCTURES

Introduction

In this chapter we fix an o-minimal expansion $(R, <, \mathcal{S})$ of an ordered abelian group $(R, <, 0, -, +)$. (As was shown in Chapter 1, this group operation makes R an abelian divisible torsion-free group, so we may and do consider R as a vector space over $\mathbb{Q}$.)

For convenience we fix a positive element $1 \in R$. We also set

$$|x| := \begin{cases} x \text{ if } x \geq 0, \\ -x \text{ if } x < 0. \end{cases}$$

The presence of such a group operation leads in Section 1 to quick proofs of curve selection and of the fact that a definable continuous map on a closed bounded definable set has closed bounded image.

Definable curves here play a role similar to that of sequences in $\mathbb{R}^n$, but have better properties. Section 2 proves that "fiberwise open" implies "piecewise open" for definable sets $S \subseteq R^{m+n}$, and variants of this important fact. In Section 3 we show that "definably connected" equals "definably path connected", and we obtain some results on definable partitions of unity. In Section 4 we consider definably proper and definably identifying maps.

§1. Curve selection

(1.1) An important consequence of the presence of the group structure is that we can definably pick an element $e(X) \in X$ from each nonempty definable set X. The idea is to let $e(X)$ be the midpoint of X if X is a bounded interval. Here are the details:

(i) Let $X \subseteq R$ be definable and nonempty. If X has a least element, then we let $e(X)$ be this least element. If X does not have a least element, let (a, b) be its "left-most" interval: $a = \inf X$, $b = \sup\{x \in R : (a, x) \subseteq R\}$. Then

$a < b$ and $(a, b) \subseteq X$; now put

$$e(X) := \begin{cases} 0 & \text{if } a = -\infty, b = +\infty, \\ b - 1 & \text{if } a = -\infty,\ b \in R, \\ a + 1 & \text{if } a \in R,\ b = +\infty, \\ (a+b)/2 & \text{if } a, b \in R. \end{cases}$$

(ii) Let $X \subseteq R^m$ be definable and nonempty, $m > 1$, and let $\pi : R^m \to R^{m-1}$ be the projection on the first $m-1$ coordinates. Then $\pi X \subseteq R^{m-1}$ so we may assume inductively that an element $a = e(\pi X)$ of πX has been defined. Then $X_a \subseteq R$ and we put $e(X) := \big(a, e(X_a)\big)$.

(1.2) PROPOSITION (DEFINABLE CHOICE).

(i) *If $S \subseteq R^{m+n}$ is definable and $\pi : R^{m+n} \to R^m$ the projection on the first m coordinates, then there is a definable map $f : \pi S \to R^n$ such that $\Gamma(f) \subseteq S$.*

(ii) *Each definable equivalence relation on a definable set X has a definable set of representatives.*

PROOF. For (i), define $f(x) := e(S_x)$ for $x \in \pi S$. For (ii) note that

$$\{e(A) : \ A \text{ is an equivalence class}\}$$

is a definable set of representatives. □

(1.3) COMMENT. Model-theorists will recognize in (i) the property of having definable Skolem functions. As to (ii), consider our structure $\mathcal{S}$ on R as a category with the definable sets as objects and the definable maps between them as morphisms. If $E \subseteq X \times X$ is a definable equivalence relation on a definable set $X \subseteq R^m$, then (ii) provides a definable map $f : X \to X$ such that $E = \text{kernel}(f)$, that is, $xEy \Leftrightarrow f(x) = f(y)$, for all $x, y \in X$. Hence for each definable map $g : X \to R^n$ such that $xEy \Rightarrow g(x) = g(y)$ for $x, y \in X$, there is a unique definable map $g' : f(X) \to R^n$ such that $g = g' \circ f$. Therefore the definable set $f(X)$, together with the map $f : X \to f(X)$, serves as a quotient X/E in the category $\mathcal{S}$.

(1.4) We equip from now on R^m, $m > 0$, with the "supnorm" $|\ |$:

$$|x| = \max\{|x_1|, \ldots, |x_m|\} \text{ for } x = (x_1, \ldots, x_m).$$

Note that then $|\ | : R^m \to R$ is a definable continuous function. We also define $|\ | : R^0 = \{0\} \to R$ by $|0| = 0$.

(1.5) COROLLARY (CURVE SELECTION). *If $a \in \text{cl}(X) - X$, where X is definable, then there is a definable continuous injective map $\gamma : (0, \epsilon) \to X$, for some $\epsilon > 0$, such that $\lim_{t \to 0} \gamma(t) = a$.*

PROOF. Since $a \in \text{cl}(X) - X$, the definable set $\{|a - x| : \ x \in X\} \subseteq R$ contains arbitrarily small positive elements, hence contains an interval $(0, \epsilon)$, $\epsilon > 0$. For each

t in this interval there is $x \in X$ with $|a-x| = t$. By definable choice there is then a definable map $\gamma : (0,\epsilon) \to X$ such that $|a - \gamma(t)| = t$ for all $t \in (0,\epsilon)$. By decreasing ϵ if necessary we may assume by the monotonicity theorem that γ is continuous. Obviously γ is injective and $\lim_{t\to 0} \gamma(t) = a$. $\square$

(1.6) Definable curve selection fails in the o-minimal structure $(\mathbb{R}, <)$ (see exercise 8 at the end of this section), so our standing assumption in this chapter that we are dealing with an o-minimal expansion of an ordered group is appropriate here. A point in the closure of a set X in a metric space is the limit of a sequence in X. This fact is not available in our context, but curve selection offers a *more* than adequate alternative (curves instead of sequences). We shall use curve selection in this way to prove that the image of a closed bounded definable set under a continuous definable map is closed and bounded. Here a set $A \subseteq R^m$ is called **bounded** if $|a| < r$ for all a in A and a fixed $r \in R$. First some lemmas.

(1.7) LEMMA. *Let C be a bounded cell in R^m, $m > 1$, and $\pi : R^m \to R^{m-1}$ the projection on the first $m-1$ coordinates. Then $\pi\,\mathrm{cl}(C) = \mathrm{cl}(\pi C)$.*

PROOF. We shall do the case $C = (f,g)_{\pi C}$ and leave the other case to the reader. By the continuity of π we have $\pi\,\mathrm{cl}(C) \subseteq \mathrm{cl}(\pi C)$. Let $a \in \mathrm{cl}(\pi C)$. We have to find s in R such that $(a,s) \in \mathrm{cl}(C)$. Of course we may assume $a \notin \pi C$. Then there is a continuous definable map $\gamma : (0,\epsilon) \to \pi C$ such that $\lim_{t\to 0} \gamma(t) = a$. We are going to lift the curve γ to a curve in C. Since C is bounded there is $r > 0$ such that $-r < f(x) < g(x) < r$ for all $x \in \pi C$. Define the continuous function $\lambda : (0,\epsilon) \to R$ by $\lambda(t) := \big(f(\gamma(t)) + g(\gamma(t))\big)/2$. Note that $-r < \lambda(t) < r$ for all t, so by the monotonicity theorem there is $s \in R$ such that $\lim_{t\to 0} \lambda(t) = s$. Then $t \mapsto \big(\gamma(t), \lambda(t)\big) : (0,\epsilon) \to C$ is a continuous definable map whose limit as t goes to 0 equals (a,s), so $(a,s) \in \mathrm{cl}(C)$. $\square$

REMARK. We cannot omit the hypothesis that C is bounded: in the ordered field of real numbers, consider $C := \big\{(x, 1/x) \in \mathbb{R}^2 : \ x > 0\big\}$.

(1.8) Next we recall that if $f : X \to Y$ is a continuous map from a topological space X into a Hausdorff space Y, then its graph $\Gamma(f)$ is a closed subset of $X \times Y$.

(1.9) LEMMA. *Let $f : X \to R^n$ be a definable continuous map on a closed bounded set $X \subseteq R^m$. Then $f(X)$ is bounded in R^n.*

PROOF. Suppose $\forall t \in R\ \exists x \in X\ |f(x)| > t$. By definable choice there is then a definable map $g : R \to X$ such that $\big|f\big(g(t)\big)\big| > t$ for all t in R. Since X is closed and bounded it follows from the monotonicity theorem, applied to the m coordinates of g, that $\lim_{t\to\infty} g(t) = x$ exists and belongs to X. So $f(x) = f\big(\lim_{t\to\infty} g(t)\big) = \lim_{t\to\infty} f\big(g(t)\big)$, but the last limit cannot exist in R, since $\big|f\big(g(t)\big)\big| > t$ for all t. Contradiction. $\square$

(1.10) PROPOSITION. *If $f : X \to R^n$ is a continuous definable map on a closed bounded set $X \subseteq R^m$, then $f(X)$ is closed and bounded in R^n.*

PROOF. By (1.9) we already know that $f(X)$ is bounded in R^n, so it suffices to show that $f(X)$ is closed. Consider $Y := \{(f(x), x) : x \in X\} \subseteq R^{n+m}$ (the "reversed" graph of f) and write

$$Y = C_1 \cup \cdots \cup C_k,$$

where $C_1, \ldots, C_k$ are cells in R^{n+m}. Since Y is closed by (1.8) we also have

$$Y = \mathrm{cl}(C_1) \cup \cdots \cup \mathrm{cl}(C_k).$$

By (1.9) the set Y is bounded, so the cells $C_1, \ldots, C_k$ are bounded. Then it follows from (1.7) that

$$\begin{aligned} \pi Y &= \pi\, \mathrm{cl}(C_1) \cup \cdots \cup \pi\, \mathrm{cl}(C_k) \\ &= \mathrm{cl}(\pi C_1) \cup \cdots \cup \mathrm{cl}(\pi C_k), \end{aligned}$$

where $\pi : R^{n+m} \to R^n$ is the projection on the first n coordinates. But $\pi Y = f(X)$, so $f(X)$ is closed in R^n. $\square$

Here are three easy consequences, the first dealing with the case $n = 1$.

(1.11) COROLLARY. *If $f : X \to R$ is a continuous definable function on a nonempty closed bounded set $X \subseteq R^m$, then f assumes a maximum and a minimum value.*

(1.12) COROLLARY. *If $f : X \to R^n$ is an injective continuous definable map on a closed bounded set $X \subseteq R^m$, then f is a homeomorphism from X onto $f(X)$.*

This last corollary perhaps requires explanation: f maps *definable* closed subsets of X onto closed subsets of $f(X)$, by (1.10), hence it maps *definable* open subsets of X onto open subsets of $f(X)$; now use the fact that an arbitrary open subset of X is a union of definable open subsets of X.

(1.13) COROLLARY. *Let $f : X \to R^n$ be a definable continuous map on a closed bounded set $X \subseteq R^m$ and let $Y = f(X)$. Then we have:*

(i) *A definable set $S \subseteq Y$ is closed if and only if $f^{-1}(S)$ is closed;*
(ii) *A definable map $g : Y \to R^p$ is continuous if and only if $g \circ f : X \to R^p$ is continuous.*

PROOF. Assertion (i) is clear from (1.10) and (ii) is an easy consequence of (i), taking into account that for continuity of g it suffices that $g^{-1}(Z)$ is closed for all definable closed $Z \subseteq R^p$. $\square$

Closed bounded definable sets have a property that is like the completeness property of projective varieties in algebraic geometry:

(1.14) PROPOSITION. *Let $X \subseteq R^m$ be a closed bounded definable set and $Y \subseteq R^n$ any definable set. Then the projection map $X \times Y \to Y$ maps definable closed subsets of $X \times Y$ onto definable closed subsets of Y.*

PROOF. Let $q: X \times Y \to Y$ be the projection map, let $A \subseteq X \times Y$ be definable and closed in $X \times Y$, and suppose $y \in \mathrm{cl}_Y\big(q(A)\big)$. Then there is for each $t > 0$ in R a point $a \in A$ such that $|q(a) - y| < t$. By definable choice and the monotonicity theorem this gives a definable continuous map $\alpha : (0, \epsilon) \to A$, for some $\epsilon > 0$, such that $\big|q\big(\alpha(t)\big) - y\big| < t$ for $t \in (0, \epsilon)$. Write $\alpha(t) = \big(\beta(t), \gamma(t)\big)$, where β and $\gamma = q \circ \alpha$ are definable continuous maps from $(0, \epsilon)$ into X and Y respectively, so $\lim_{t \to 0} \gamma(t) = y$. Since X is bounded, it follows from Chapter 3, (1.6) that $x := \lim_{t \to 0} \beta(t)$ exists in R^m; hence $x \in X$, since X is closed. Then $(x, y) \in X \times Y$ and $\lim_{t \to 0} \alpha(t) = (x, y)$, so $(x, y) \in A$, because A is closed in $X \times Y$. Therefore $y = q(x, y) \in q(A)$ as desired. □

(1.15) EXERCISES.

In the first three exercises we consider a definable equivalence relation $E \subseteq X \times X$ on a definable set $X \subseteq R^m$.

1. Show that the definable set of representatives indicated in the proof of (1.2) is definable in the model-theoretic structure $(R, <, 1, +, E)$. (Recall that this set of representatives is given by $T := \big\{e(A) : \ A \text{ is an equivalence class}\big\}$.)

2. Show that E has only finitely many equivalence classes of dimension $\dim(X)$, and that each of them is definable in the model-theoretic structure $(R, <, 1, +, E)$.

3. Suppose all equivalence classes of E have the same Euler characteristic e. Show that then the Euler characteristic of X is a multiple of e. (In particular, this shows that for $e > 1$ there is no definable equivalence relation on R^m all of whose equivalence classes have exactly e elements.)

4. (Uniform continuity) Let $X \subseteq R^m$ be a closed and bounded definable set and $f: X \to R^n$ a continuous definable map. Show that there is for each $\epsilon > 0$ a $\delta > 0$ such that whenever $|x - y| < \delta$, $x, y \in X$, we have $|f(x) - f(y)| < \epsilon$.

5. (Fixed point theorem) Let X be a nonempty closed bounded definable subset of R^m and $f: X \to X$ a definable map such that $|f(x) - f(y)| < |x - y|$ for all distinct points $x, y \in X$. Show that f has a unique fixed point.

6. (Uniform curve selection) Let $X \subseteq R^m$ be definable. Show there are definable maps $\epsilon : \partial X \to (0, \infty)$ and $\Gamma : (0, \epsilon) \to X$ such that for each $a \in \partial X$ the function $t \to \Gamma(a, t) : \big(0, \epsilon(a)\big) \to X$ is continuous, injective and satisfies $\lim_{t \to 0} \Gamma(a, t) = a$.

7. Given a map $f : A \to R^n$, $A \subseteq R^m$, we call f **locally bounded** if each point $a \in A$ has a neighborhood U in A such that $f(U)$ is bounded. Let $A \subseteq R^m$ be definable and $f: A \to R^n$ definable. Prove the following equivalence:

$$f \text{ is continuous} \iff f \text{ is locally bounded and } \Gamma(f) \text{ is closed in } A \times R^n .$$

Note that the local boundedness condition cannot be omitted in the reverse implication $\Leftarrow$: the function $f:\mathbb{R}\to\mathbb{R}$ given by $f(x)=1/x$ for $x\neq 0$, $f(0)=0$, is definable in the ordered field of real numbers, has closed graph $\Gamma(f)$ in $\mathbb{R}^2$, but is not continuous at 0.

8. Consider the o-minimal model-theoretic structure $(\mathbb{R},<)$ and the set

$$X := \{(x,y)\in\mathbb{R}^2 :\ 0<x<1,\ 0<y<2\},$$

which is definable in $(\mathbb{R},<)$ using the constants $0,1,2$. Note that $(1,2)\in \mathrm{cl}(X)$ and show that there is no subset Y of X such that Y is definable in $(\mathbb{R},<)$ using constants, $\dim(Y)=1$ and $(1,2)\in\mathrm{cl}(Y)$.

§2. Fiberwise properties

(2.1) Strictly speaking the results of this section are not needed later in this book. However, for those familiar with elementary model theory this section provides a very cheap way to obtain the triviality theorem of Chapter 9 from the triangulation theorem of Chapter 8, see Chapter 8, (2.13) and (2.14), for more details. The proof of the theorem below nicely illustrates the use of the definable choice principle.

(2.2) THEOREM. *Let $S\subseteq R^{m+n}$ be a definable set such that for each $x\in R^m$ the fiber S_x is open in R^n. Then there is a partition of R^m into cells $C_1,\dots,C_k$ such that $S\cap(C_i\times R^n)$ is open in $C_i\times R^n$ for $i=1,\dots,k$. (Same for "closed" instead of "open".)*

PROOF. By induction on m. The case $m=0$ is trivial. Let $m>0$ and assume the result holds for lower values of m. First we show:

$$(*)\qquad \begin{cases}\text{Each open cell } C \text{ in } R^m \text{ has an open subcell } D\\ \text{such that } S\cap(D\times R^n) \text{ is open in } D\times R^n.\end{cases}$$

To see this, note that $C=C(1)\cup C(2)$ with definable $C(i)$ as follows:

$$\begin{aligned} C(1) &:= \{x\in C:\ (\{x\}\times S_x)\cap \mathrm{cl}((C\times R^n)-S)\neq\emptyset\},\\ C(2) &:= \{x\in C:\ (\{x\}\times S_x)\cap \mathrm{cl}((C\times R^n)-S)=\emptyset\}.\end{aligned}$$

If $C(2)$ has nonempty interior, we can take for D any open cell contained in $C(2)$. Now assume $C(2)$ has empty interior. Then $C(1)$ has nonempty interior, and replacing C by an open subcell contained in $C(1)$ we reduce to the case that $C=C(1)$. So for each $x\in C$ there is a point $s(x)\in S_x$ with $(x,s(x))\in\mathrm{cl}((C\times R^n)-S)$. Since S_x is open in R^n there is also for each $x\in C$ an $\epsilon(x)>0$ such that if $y\in R^n$ and $|y-s(x)|<\epsilon(x)$, then $y\in S_x$. By definable choice we may take $x\mapsto s(x):C\to R^n$ and $x\mapsto\epsilon(x):C\to R$ to be definable functions: replacing C by a suitable open subcell we may further reduce to the case that these functions are continuous.

Then $\{(x,y) \in C \times R^n : x \in C, |y - s(x)| < \epsilon(x)\}$ is a subset of S, and open in $C \times R^n$; it also contains the points $(x, s(x))$ $(x \in C)$, which belong to the closure of $(C \times R^n) - S$, a blatant contradiction. This proves $(*)$.

Now let A be the union of all open boxes D in R^m such that $S \cap (D \times R^n)$ is open in $D \times R^n$. Then A is definable, $S \cap (A \times R^n)$ is open in $A \times R^n$, and $(*)$ implies that $\dim(R^m - A) < m$. Write $R^m - A$ as a disjoint union of cells $B_1, \ldots, B_r$ where each B_i has dimension $< m$.

Let B be one of the cells B_i, and let $\dim(B) = d < m$. Then $p(B)$ is an open cell in R^d, and we can use the homeomorphism $(x,y) \mapsto (p_B(x), y) : B \times R^n \to p(B) \times R^n$ and apply the inductive hypothesis to the image of $S \cap (B \times R^n)$ under this homeomorphism. The "closed" version follows by passing to complements. □

(2.3) COROLLARY. *Let $S' \subseteq S$ be definable sets in R^{m+n}, and let $A \subseteq R^m$ be definable such that S'_a is open in S_a for all $a \in A$. Then there is a partition of A into definable subsets $A_1, \ldots, A_M$ such that $S' \cap (A_i \times R^n)$ is open in $S \cap (A_i \times R^n)$, for $i = 1, \ldots, M$. (Same with "closed" instead of "open".)*

PROOF. In this case it is more convenient to do the closed version. Replacing S by $S \cap (A \times R^n)$ we may as well assume that the projection map $R^{m+n} \to R^m$ maps S into A; thus S'_x is closed in S_x for each $x \in R^m$. Let S^* be the subset of R^{m+n} with $S^*_x = \mathrm{cl}(S'_x)$ for each $x \in R^m$. Then S^* is definable and $S^*_x \cap S_x = S'_x$ for $x \in R^m$. Now apply the closed version of the theorem above to S^*. □

(2.4) COROLLARY. *Let $S \subseteq R^{m+n}$ be definable, $f : S \to R^k$ a locally bounded definable map, and $A \subseteq R^m$ a definable set such that for all $a \in A$ the map $f_a : S_a \to R^k$ is continuous, where $f_a(y) := f(a,y)$. Then there is a partition of A into definable subsets $A_1, \ldots, A_M$ such that each restriction*

$$f|S \cap (A_i \times R^n) : S \cap (A_i \times R^n) \to R^k$$

is continuous.

PROOF. Apply the closed version of (2.3) to the definable subset $\Gamma(f)$ of $S \times R^k$, noting that $\Gamma(f)_a = \Gamma(f_a)$ is closed in $S_a \times R^k = (S \times R^k)_a$ for each $a \in A$. This gives a partition of A into definable subsets $A_1, \ldots, A_M$ such that for each $i \in \{1, \ldots, M\}$ the set $\Gamma(f) \cap (A_i \times R^{n+k}) = \Gamma(f|S \cap (A_i \times R^n))$ is closed in $(S \times R^k) \cap (A_i \times R^{n+k}) = (S \cap (A_i \times R^n)) \times R^k$. Hence, by (1.15), exercise 7, the restrictions $f|S \cap (A_i \times R^n)$ are continuous. □

(2.5) EXERCISES.

1. Assume $(R, <, \mathcal{S})$ expands an ordered field. Show that then (2.4) holds even without the assumption that f is locally bounded.

2. Assume $(R, <, \mathcal{S})$ expands an ordered field. Let $S \subseteq R^{m+n}$ be definable, $f: S \to R^k$ a definable map, and $A \subseteq R^m$ a definable set such that $f|S \cap (A \times R^n)$ is injective and $f_a: S_a \to R^k$ is a homeomorphism from S_a onto $f_a(S_a)$ for all $a \in A$. Show that there is a partition of A into definable subsets $A_1, \ldots, A_M$ such that each restriction $f|S \cap (A_i \times R^n): S \cap (A_i \times R^n) \to R^k$ is a homeomorphism from $S \cap (A_i \times R^n)$ onto $f\big(S \cap (A_i \times R^n)\big)$.

§3. Paths and partitions of unity

The two topics of this section are unrelated. The results here will be essential in several later chapters.

(3.1) DEFINABLE PATHS.

Let $X \subseteq R^m$. A **definable path** in X is a definable continuous map $\gamma: [a, b] \to R^m$ with $a, b \in R$, $a < b$, taking values in X. Such a path γ is said to connect the points $\gamma(a)$ and $\gamma(b)$. If $\gamma: [a, b] \to X$ and $\delta: [b, c] \to X$ are definable paths in X with $\gamma(b) = \delta(b)$ we have a definable path $\gamma \vee \delta: [a, c] \to X$ in X agreeing on $[a, b]$ with γ and on $[b, c]$ with δ. Using this concatenation construction it is clear that if the points $x, y \in X$ can be connected by a definable path in X, and also the points $y, z \in X$, then x, z as well. It is easy to see that if X is definable and every two points in X can be connected by a definable path in X, then X is definably connected. The converse fails for the o-minimal model-theoretic structure $(\mathbb{R}, <)$ (see the exercise at the end of this section). It does hold however under our standing assumption in this chapter that we are dealing with an o-minimal expansion of an ordered abelian group:

(3.2) PROPOSITION. *Suppose the definable set X is definably connected. Then any two points in X can be connected by a definable path in X.*

PROOF. Assume first that X is a cell, without loss of generality an open cell in R^m. This case is handled by induction on m. The case $m = 1$ is obvious. For $m > 1$, let C be the projection of X in R^{m-1}, so $X = (f, g)$ with f, g functions on C, and let (y, r) and (z, s) be two points in X, $y, z \in C$. Assume f and g are R-valued. (The case that $f = -\infty$ or $g = +\infty$ is left to the reader.)

We first connect (y, r) by a "vertical" path in X to $(y, (f(y) + g(y))/2)$; by the inductive hypothesis there is a definable path $\gamma: [a, b] \to C$ connecting y to z and this path lifts to the path $[a, b] \to X$ given by $t \to \big(\gamma(t), \big(f(\gamma(t)\big) + g(\gamma(t))\big)/2\big)$, connecting $\big(y, \big(f(y) + g(y)\big)/2\big)$ to $\big(z, \big(f(z) + g(z)\big)/2\big)$; the last point can in turn be connected to (z, s) by a vertical path in X. Concatenating the three paths we obtain a definable path in X connecting (y, r) to (z, s).

In the general case, we use Chapter 3, (2.19), exercise 5 to write X as a union of cells $C_1, \ldots, C_k$ where for each $i < k$, either C_i intersects the closure of C_{i+1}, or

C_{i+1} intersects the closure of C_i. Then by curve selection there is a definable path in $C_i \cup C_{i+1}$ that connects a point of C_i with a point in C_{i+1}. Now combine this with the fact that the desired result has already been proved for cells. $\square$

One particular consequence of (3.2) will be used in Chapters 8 and 9 in proofs of triangulation and trivialization:

(3.3) COROLLARY. *Let X and Y be definable sets in R^n with X definably connected, and $X \cap \mathrm{bd}(Y) = \emptyset$. Then either $X \subseteq Y$ or $X \cap Y = \emptyset$.*

PROOF. If not, there would be points $x \in X - Y$ and $y \in X \cap Y$. Take a definable path $\gamma : [a, b] \to X$ connecting x to y. Put $c := \inf\{t \in [a, b] : \gamma(t) \in Y\}$. Then $\gamma(c) \in X \cap \mathrm{bd}(Y)$, contradiction. $\square$

PARTITIONS OF UNITY.

(3.4) LEMMA. *Let $A \subseteq B \subseteq R^m$ be definable sets, with A closed in B. Then there is a definable continuous function $f : B \to [0, 1]$ with $f^{-1}(0) = A$.*

PROOF. Assuming $A \neq \emptyset$, take $f(x) = \min(1, d(x, A))$, where

$$d(x, A) = \inf\{|x - a| : a \in A\}. \quad \square$$

(3.5) LEMMA. *Let A_0 and A_1 be disjoint definable closed subsets of the definable set $B \subseteq R^m$. Then there are disjoint definable open subsets U_0 and U_1 of B with $A_i \subseteq U_i$, $i = 0, 1$.*

PROOF. Take functions f_0 and f_1 as in (3.4), and put

$$U_0 := \{x \in B : f_0(x) < f_1(x)\}, \text{ and } U_1 := \{x \in B : f_1(x) < f_0(x)\}. \quad \square$$

(3.6) LEMMA (SHRINKING OF OPEN COVERINGS). *Let the definable set $B \subseteq R^m$ be the union of its definable open subsets $U_1, \ldots, U_n$. Then B is also the union of definable open subsets $V_1, \ldots, V_n$ with $\mathrm{cl}_B(V_i) \subseteq U_i$ for $i = 1, \ldots, n$.*

PROOF. Assume inductively that $V_i \subseteq U_i$ has been defined for $i = 1, \ldots, k$ $(k < n)$ such that V_i is definable, open in B, $\mathrm{cl}_B(V_i) \subseteq U_i$ and $V_1, \ldots, V_k, U_{k+1}, \ldots, U_n$ cover B. Then apply (3.5) to the following two disjoint closed subsets of B:

$$A_0 := B - U_{k+1},$$
$$A_1 := B - \left(\bigcup_{i=1}^{k} V_i \cup \bigcup_{j=k+2}^{n} U_j \right). \quad \square$$

(3.7) Lemma (definable partition of unity). *With $U_1, \dots, U_n$ and B as in (3.6), there are definable continuous functions $f_1, \dots, f_n : B \to [0,1]$ such that*

(i) $\operatorname{supp}(f_i) \subseteq U_i$ *for* $i = 1, \dots, n$, *where* $\operatorname{supp}(f_i) := \operatorname{cl}_B(\{x \in B :\ f_i(x) \neq 0\})$,
(ii) $\sum f_i(x) > 0$ *for all* $x \in B$.

Moreover, if $(R, <, \mathcal{S})$ *expands a real closed ordered field, then* (ii) *can be replaced by* $\sum f_i = 1$. (*We then call* $f_1, \dots, f_n$ *a definable partition of unity for the covering* $U_1, \dots, U_n$.)

The easy proof using (3.6) and (3.4) is left to the reader.

(3.8) Lemma. *Suppose* $(R, <, \mathcal{S})$ *expands an ordered real closed field. Let* A_0 *and* A_1 *be disjoint definable closed subsets of a definable set* $B \subseteq R^m$. *Then there is a continuous definable function* $f : B \to [0,1]$ *with* $f^{-1}(0) = A_0$ *and* $f^{-1}(1) = A_1$.

Proof. Choose U_0 and U_1 as in (3.5), and apply (3.7) to get a definable partition of unity f_0, f_1, f_2 for the covering $U_0, U_1, B - (A_0 \cup A_1)$ of B. Then use (3.4) to get definable continuous functions $g_0 : B \to [0, 1/2]$ and $g_1 : B \to [1/2, 1]$ with $g_0^{-1}(0) = A_0$, $g_1^{-1}(1) = A_1$. Now set $f := f_0 g_0 + f_1 g_1 + (1/2) f_2$. $\square$

(3.9) Exercise. Show that in the model-theoretic structure $(\mathbb{R}, <, 0, 1, 2)$ the definable set $\{(x,y) :\ 0 < x < 1,\ 0 < y < 2\} \cup \{(1,2)\}$ in $\mathbb{R}^2$ is definably connected but not definably path connected (the latter defined as in (3.1)).

§4. Curves, proper maps, and identifying maps

This section contains miscellaneous results used only in the last chapter of this book. The reason for including these results here is simply that they hold under weaker assumptions than are in force later on in this book.

(4.1) Definitions. Let $X \subseteq R^m$ be a definable set.

(1) A **definable curve in** X is a definable map $\gamma : I \to X$, for some interval $I = (a, b)$ in R. We do not require γ to be continuous, and we allow of course that $b = +\infty$. Our interest is in the behavior of γ near its right endpoint b, that is in the germ of γ at b; we note that γ will be continuous on some smaller interval (a', b) with the same right endpoint. (That we are only interested in the behavior near the right endpoint is just a convention. We could as well have chosen the left endpoint.)
(2) Let $\gamma : (a, b) \to X$ be a definable curve in X. Given a point $p \in R^m$ we write $\gamma \to p$ as shorthand notation for $\lim_{t \to b} \gamma(t) = p$. (We do not require that $p \in X$.)

Let $X \subseteq R^m$ be definable and γ a definable curve in X. We call γ **completable** if there is a (necessarily unique) point $p \in R^m$ such that $\gamma \to p$. If $\gamma \to p \in X$, then we call γ **completable in** X. Here are some simple observations.

(i) X bounded $\Rightarrow$ γ is completable. (By Chapter 3, (1.6).)
(ii) X is closed and bounded $\Rightarrow$ γ is completable in X.
(iii) If $f: X \to Y$ is a definable map into a definable set $Y \subseteq R^n$, then $f(\gamma) := f \circ \gamma$ is a definable curve in Y; if f is in addition surjective ($f(X) = Y$), then each definable curve β in Y can be lifted to a definable curve α in X, that is, $f(\alpha) = \beta$. (By definable selection.)
(iv) γ is either injective on a subinterval (a', b), or constant on such a subinterval. (By the monotonicity theorem.)

(4.2) LEMMA. *Let $f: X \to Y$ be a definable map from the definable set $X \subseteq R^m$ into the definable set $Y \subseteq R^n$, and let $p \in X$. Then f is continuous at the point p iff for each definable curve γ in X with $\gamma \to p$ we have $f(\gamma) \to f(p)$.*

PROOF. If f is continuous at p, the "curve continuity" follows immediately. Suppose f is not continuous at p; so there is $\epsilon > 0$ such that the definable set $\{|x - p| : x \in X, |f(x) - f(p)| \geq \epsilon\}$ contains arbitrarily small positive elements, so it contains an interval I with left endpoint 0. By definable selection this produces a definable curve $\gamma: I \to X$ such that $|\gamma(t) - p| = t$ and $|f(\gamma(t)) - f(p)| \geq \epsilon$ for $t \in I$. Then its "transform" $\gamma': -I \to X$ given by $\gamma'(-t) = \gamma(t)$ has the property that $\gamma' \to p$ but not $f(\gamma') \to f(p)$. □

REMARK. As the proof shows, it suffices to consider just injective definable curves γ in the statement of the lemma.

(4.3) LEMMA. *Let the definable curve $\beta: (a, b) \to R^n$ be completable, $\beta \to p$, and let $\epsilon > 0$ be such that $a < b - \epsilon < b$ and β is continuous on $[b - \epsilon, b)$. Suppose $\alpha: I \to \beta([b - \epsilon, b))$ is a definable curve in $\beta([b - \epsilon, b))$ not completable in $\beta([b - \epsilon, b))$. Then α is a reparametrization of β near b: there are a subinterval I' of I with the same right endpoint as I, an $a' \in (b - \epsilon, b)$ and a definable strictly increasing homeomorphism $h: I' \to (a', b)$ such that $\alpha(t) = \beta(h(t))$ for $t \in I'$.*

PROOF. The curve α is necessarily completable in the closed and bounded set $\beta([b - \epsilon, b)) \cup \{p\}$. Hence $\alpha \to p$, and α and β are both injective near b. Thus by decreasing the interval I, keeping the same right endpoint, and also decreasing ϵ, we may as well assume that α is continuous and injective, and β is injective on $[b - \epsilon, b)$, and $p \notin \beta([b - \epsilon, b))$. Write $\alpha = (\alpha_1, \ldots, \alpha_n)$ and $\beta = (\beta_1, \ldots, \beta_n)$. By further shrinking we may also assume one of the β_i is strictly monotone on $[b - \epsilon, b)$, say β_1 is strictly increasing on $[b - \epsilon, b)$. Since $\alpha_1(I) \subseteq \beta_1([b - \epsilon, b))$, the function α_1 is also strictly increasing near the right endpoint of I.

Now use exercise 2 from Chapter 3, (1.9) to conclude that there are a subinterval I' of I with the same right endpoint as I, and $a' \in (b - \epsilon, b)$ and a definable strictly increasing homeomorphism $h: I' \to (a', b)$ such that $\alpha_1(t) = \beta_1(h(t))$ for $t \in I'$. Then it follows easily from $\alpha(I) \subseteq \beta([b - \epsilon, b))$ and the injectivity of β_1 on $[b - \epsilon, b)$ that also $\alpha(t) = \beta(h(t))$ for $t \in I'$. □

(4.4) DEFINITION. Let $f: X \to Y$ be a definable continuous map from the definable set $X \subseteq R^m$ into the definable set $Y \subseteq R^n$.

(1) We call f **definably proper** if for each definable set $K \subseteq Y$ we have: K is closed and bounded in $R^n \Rightarrow f^{-1}(K) \subseteq X$ is closed and bounded in R^m.
(2) We call f **definably identifying** if f is surjective ($f(X) = Y$), and for each definable set $K \subseteq Y$ we have: $f^{-1}(K)$ is closed in $X \Rightarrow K$ is closed in Y.

If $X \subseteq R^m$ is a closed and bounded definable set and $f: X \to R^n$ is a definable continuous map, then f is obviously definably proper, and, as a map from X onto $f(X)$, also definably identifying, by (1.10). A more typical example of a definably proper map is a projection map $R^m \times Y \to R^m$, where $Y \subseteq R^n$ is a closed and bounded definable set. These notions, and the lemmas that follow, will become especially useful in the last chapter when we construct quotient spaces of definable sets modulo definable equivalence relations.

(4.5) LEMMA. *Let $f: X \to Y$ be as above: X, Y definable, f definable and continuous. Then*

(1) *f is definably proper iff each definable curve γ in X whose image $f(\gamma)$ is completable in Y is completable in X;*
(2) *f is definably identifying iff each definable curve β in Y that is completable in Y lifts to a definable curve α in X that is completable in X.*

PROOF. (1) Suppose f is definably proper, $\gamma: (a, b) \to X$ is a definable curve in X and $f(\gamma)$ is completable in Y, say $f(\gamma) \to y \in Y$. Take $a' \in (a, b)$ such that γ is continuous on $[a', b)$. Then the definable set

$$K := \{f(\gamma(t)) : t \in [a', b)\} \cup \{y\} \subseteq Y$$

is closed and bounded in R^n, so $f^{-1}(K) \subseteq X$ is closed and bounded in R^m, so γ must be completable in $f^{-1}(K)$, hence in X.

Conversely, suppose f is not definably proper; then there is a definable $K \subseteq Y$ that is closed and bounded in R^n, such that $f^{-1}(K)$ is not closed in R^m or not bounded. If $f^{-1}(K)$ is not bounded, we can by definable selection choose a definable curve γ in $f^{-1}(K)$ that is not completable in R^m. Then $f(\gamma)$ is a curve in K, hence completable in K, and so in Y. If $f^{-1}(K)$ is not closed in R^m, take $p \in \mathrm{cl}(f^{-1}(K)) - f^{-1}(K)$, and use curve selection to get a definable curve γ in $f^{-1}(K)$ such that $\gamma \to p$. Since $f^{-1}(K)$ is closed in X, we have $p \notin X$, so γ is not completable in X. But, as before, $f(\gamma)$ is completable in Y.

(2) Suppose f is definably identifying, and let $\beta: (a, b) \to Y$ be a definable curve in Y that is completable in Y. We have to lift β to a curve α in X that is completable in X. We may assume β is injective and continuous on a subset $[b - \epsilon, b)$ of (a, b), $\epsilon > 0$, since otherwise β would be constant on such a set, and we can then use the fact that f is surjective.

After further decreasing ϵ we may also assume that $\beta \to y \in Y - \beta([b-\epsilon, b))$, so $\beta([b-\epsilon, b))$ is not closed in Y. Hence $f^{-1}(\beta([b-\epsilon, b)))$ is not closed in X. By curve selection there is a definable curve α in $f^{-1}(\beta([b-\epsilon, b)))$ with $\alpha \to x$ and $x \in X - f^{-1}(\beta([b-\epsilon, b)))$. Then $f(\alpha)$ is a definable curve in $\beta([b-\epsilon, b))$ with $f(\alpha) \to f(x)$ and $f(x) \notin \beta([b-\epsilon, b))$. Hence by lemma (4.3) above, $f(\alpha)$ is a reparametrization of β near b. Reparametrizing α we achieve that $f(\alpha) = \beta$.

Conversely, suppose f is surjective but not definably identifying. Let $K \subseteq Y$ be definable such that $f^{-1}(K)$ is closed in X but K is not closed in Y. Then there is a definable curve β in K with $\beta \to y \in Y - K$. If β were to lift to a definable curve α in X that is completable in X, say $\alpha \to x \in X$, then image$(\alpha) \subseteq f^{-1}(K)$ and $f^{-1}(K)$ would be closed in X, so $x \in f^{-1}(K)$, hence $y = f(x) \in K$, contradiction. $\square$

(4.6) COROLLARY. *If $f: X \to Y$ as above is definably proper and surjective, then f is definably identifying.*

This is immediate from (4.5).

(4.7) COROLLARY. *Let $f: X \to Y$ as above be surjective. Then f is definably identifying iff for each definable $K \subseteq Y$ we have*

$$f(\mathrm{cl}_X(f^{-1}(K))) = \mathrm{cl}_Y(K).$$

PROOF. Suppose f is definably identifying, and let $K \subseteq Y$ be definable. By continuity of f we have $f(\mathrm{cl}_X(f^{-1}(K))) \subseteq \mathrm{cl}_Y(K)$. For the reverse inclusion, let y be in the closure of K in Y, and take a definable curve β in K with $\beta \to y$. Lift β to a definable curve α in X with $\alpha \to x \in X$, so that $f(x) = y$. Then α lies in $f^{-1}(K)$, so $x \in \mathrm{cl}_X(f^{-1}(K))$, hence $y = f(x) \in f(\mathrm{cl}_X(f^{-1}(K)))$.

Conversely, assume each definable $K \subseteq Y$ has the property mentioned in the corollary. Let $\beta: (a, b) \to Y$ be a definable curve such that $\beta \to y \in Y$. We have to lift β to a definable curve in X completable in X. We may assume that β is not constant near b, hence we may assume β is injective and continuous on a subset $[b-\epsilon, b)$ of (a, b), $\epsilon > 0$, and $y \notin K := \beta([b-\epsilon, b))$. Then $y \in \mathrm{cl}_Y(K)$, so there is an x in $\mathrm{cl}_X(f^{-1}(K))$ with $f(x) = y$. Let α be a definable curve in $f^{-1}(K)$ with $\alpha \to x$. Then $f(\alpha)$ is a definable curve in K, hence the curve $f(\alpha)$ is a reparametrization of β near b, by (4.3). Then α can be similarly reparametrized so that $f(\alpha) = \beta$. $\square$

(4.8) EXERCISES.

1. Let $f: X \to Y$ be a definably proper map. (This includes the assumption that X and Y are definable sets and f is definable and continuous.) Show that if $A \subseteq X$ is definable and closed in X, then $f(A)$ is closed in Y. Show that if $f': X' \to Y'$ is a definably proper map, then $f \times f': X \times X' \to Y \times Y'$ is definably proper. Show that if $g: Y \to Z$ is definably proper, then $g \circ f: X \to Z$ is definably proper.

2. Let $(R, <, \mathcal{S}')$ be an o-minimal structure with $\mathcal{S} \subseteq \mathcal{S}'$ and let $f : X \to Y$ be a definable continuous map between definable sets X and Y in R^m and R^n, where "definable" is taken in the sense of $\mathcal{S}$. Assume also that Y is locally closed in R^n. Show

$$f \text{ is definably proper with respect to } (R, <, \mathcal{S})$$
$$\Leftrightarrow$$
$$f \text{ is definably proper with respect to } (R, <, \mathcal{S}').$$

In the next exercise we assume that $(R, <, \mathcal{S}) = (\mathbb{R}, <, \mathcal{S})$ is an o-minimal expansion of the ordered additive group of reals. We recall that a continuous map $f : X \to Y$ between Hausdorff spaces X and Y is called **proper** if f maps closed subsets of X onto closed subsets of Y and each point $y \in Y$ has compact fiber $f^{-1}(y)$. This definition agrees with that of Bourbaki [6, Ch. 1, §10, Thm. 1] who also shows that if X and Y are Hausdorff spaces and the continuous map $f : X \to Y$ is proper, then $f^{-1}(C)$ is compact for each compact $C \subseteq Y$.

3. Let $f : X \to Y$ be a definable continuous map between definable sets $X \subseteq \mathbb{R}^m$ and $Y \subseteq \mathbb{R}^n$. Show that f is proper if and only if f is definably proper.

Notes and comments

The definable choice principle of Section 1 combines the model-theoretic properties of "definability of Skolem functions" and "elimination of imaginaries". The first use of definability of Skolem functions in the semialgebraic case seems to be in Delzell [14]. The stronger principle of definable choice was pointed out for real closed fields in [18]. That definability of Skolem function implies curve selection (for real and p-adic fields) was observed by Scowcroft and Van den Dries in [52]. Brumfiel makes the heuristically useful remark in [8] that curves play in semialgebraic geometry the same role as sequences in metric spaces. Peterzil and Steinhorn show in [47] that proposition (1.10) on images of closed bounded definable sets remains true for arbitrary o-minimal structures.

Bochnak, Coste, and Roy prove the fiberwise properties of Section 2 in the semialgebraic case in their book [4] by methods that seem specific to that situation. After the more general theorem (2.2) above had been found, Speissegger [57] established these properties for arbitrary o-minimal structures. The "partition of unity results" of Section 3 are taken over from the semialgebraic case treated by Delfs and Knebusch in [13, §1]. Similarly I adapted from Brumfiel [8] the material in Section 4 on curves, proper and identifying maps.